

- Consider $g: \mathbb{R}_2[x] \xrightarrow{\text{linear}} \mathbb{R}_1[x]$

$$\begin{array}{ccc} 1+x & \longmapsto & 1 \\ 1+x^2 & \longmapsto & 1+x \\ x+x^2 & \longmapsto & x \end{array}$$

We want to write $M_{B_1, B_2}(g)$ where $B_1 = \{1+x+x^2, 1+x^2, x+x^2\}$
 $B_2 = \{1+x, 1-x\}$

As always: $M_{B_1, B_2}(g) = \left[g(1+x+x^2)_{B_2} \mid g(1+x^2)_{B_2} \mid g(x+x^2)_{B_2} \right]$

But we don't even know $g(1+x+x^2)$.

We will use linearity of g . In fact: we know where $1+x$, $1+x^2$ and $x+x^2$ are sent to. If we write $1+x+x^2$ as lin. comb. of $1+x$, $1+x^2$ and $x+x^2$, then

$$g(1+x+x^2) = g(a(1+x) + b(1+x^2) + c(x+x^2))$$

$$\text{linearity} \quad = a g(1+x) + b g(1+x^2) + c g(x+x^2)$$

$$= a + b(1+x) + cx$$

$$= a+b + (b+c)x$$

Let's find (a, b, c) .

$$\begin{aligned}
 1+x+x^2 &= a(1+x) + b(1+x^2) + c(x+x^2) \\
 &= a+ax+b+bx^2+cx+cx^2 \\
 &= (a+b) + (a+c)x + (b+c)x^2
 \end{aligned}$$

$$\text{iff } \begin{cases} a+b=1 \\ a+c=1 \\ b+c=1 \end{cases} \xRightarrow{\text{solve it!}} \begin{cases} a=1/2 \\ b=1/2 \\ c=1/2 \end{cases}$$

$$\Rightarrow g(1+x+x^2) = a+b+(b+c)x = 1+x$$

$$\begin{aligned}
 \text{Finally, } g(1+x+x^2)_{\mathcal{B}_2} &= (1+x)_{\mathcal{B}_2} \\
 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad [\text{Since } \mathcal{B}_2 = \{1+x, 1-x\}]
 \end{aligned}$$

First column: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Do yourself $g(1+x^2)_{\mathcal{B}_2}$ and $g(x+x^2)_{\mathcal{B}_2}$.

$$\begin{aligned}
 \text{You will find } M_{\mathcal{B}_1, \mathcal{B}_2}(g) &= \left[g(1+x+x^2)_{\mathcal{B}_2} \mid g(1+x^2)_{\mathcal{B}_2} \mid g(x+x^2)_{\mathcal{B}_2} \right] \\
 &= \begin{bmatrix} 1 & 1 & 1/2 \\ 0 & 0 & -1/2 \end{bmatrix}
 \end{aligned}$$

- Consider $g: \mathbb{R}_2[x] \xrightarrow{\text{linear}} \mathbb{R}_1[x]$

$$\begin{array}{ccc} 1+x & \longmapsto & 1 \\ 1+x^2 & \longmapsto & 1+x \\ x+x^2 & \longmapsto & x \end{array}$$

We want to write $M_{e_1, e_2}(g)$ where $e_1 = \{1, x, x^2\}$
 $e_2 = \{1, x\}$

As always: $M_{e_1, e_2}(g) = [g(1)e_2 \mid g(x)e_2 \mid g(x^2)e_2]$

But we don't even know $g(1)$.

We will use linearity of g . In fact: we know where $1+x$, $1+x^2$ and $x+x^2$ are sent to. If we write 1 as lin. comb. of $1+x$, $1+x^2$ and $x+x^2$, then

$$\begin{aligned} g(1) &= g(a(1+x) + b(1+x^2) + c(x+x^2)) \\ &\stackrel{\text{linearity}}{=} a g(1+x) + b g(1+x^2) + c g(x+x^2) \\ &= a + b(1+x) + cx \\ &= a+b + (b+c)x \end{aligned}$$

Let's find (a, b, c) .

$$\begin{aligned}
 1 &= a(1+x) + b(1+x^2) + c(x+x^2) \\
 &= a + ax + b + bx^2 + cx + cx^2 \\
 &= (a+b) + (a+c)x + (b+c)x^2
 \end{aligned}$$

$$\text{iff } \begin{cases} a+b=1 \\ a+c=0 \\ b+c=0 \end{cases} \xRightarrow{\text{solve it!}} \begin{cases} a=1/2 \\ b=1/2 \\ c=-1/2 \end{cases}$$

$$\Rightarrow g(1) = a + b + (b+c)x = 1$$

$$\begin{aligned}
 \text{Finally, } g(1) &= (1)e_2 \\
 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad [\text{Since } e_2 = \{1, x\}]
 \end{aligned}$$

First column: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Do yourself $g(x)e_2$ and $g(x^2)e_2$.

$$\begin{aligned}
 \text{You will find } M_{e_1, e_2}(g) &= [g(1)e_2 \mid g(x)e_2 \mid g(x^2)e_2] \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

- $M_{e_1 e_2}(g) = M_{\mathcal{B}_2 e_2}(\text{id}) \cdot M_{\mathcal{B}_1 \mathcal{B}_2}(g) \cdot M_{e_2 \mathcal{B}_1}(\text{id})$

$$M_{\mathcal{B}_2 e_2}(\text{id}) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$M_{e_1 \mathcal{B}_1}(\text{id}) = M_{\mathcal{B}_2 e_1}(\text{id})^{-1}$$

$$M_{\mathcal{B}_1 e_2}(\text{id}) = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$